WELCOME

Pulse Production in India

Area, Production & Productivity Analysis (2016–2023)

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Course: AGR 311 (0+1) – Practical Crop Production – 1 (Rainfed) | University of Agricultural Sciences, Bangalore

Table of Contents

Overview of Presentation Structure

03 Abstract & Overview

04 Review of Literature

05 Introduction & Nutritional Importance

06 Materials & Methods

07 System Architecture & Data Pipeline

08 Flowchart – Methodology

09 Class Diagram – Pulse Data Model

10 National Pulse Trends (Results)

11 Results – Comparison with Existing System

12 Experiment & Results (Photographs)

13 Result & Discussion – Data Visualization

14 Summary & Conclusion

15 Future Scope & References

Abstract & Overview

Research Summary – Pulse Production Analysis 2016–2023

25%+	28 MT	945	#1
World Pulse Production Share	Production 2022–23	kg/ha Productivity	Global Producer

About This Research

This study comprehensively analyzes the area, production, and productivity trends of six major pulses in India — Bengal gram, pigeon pea, lentil, green gram, black gram, and horse gram — over 2016–17 to 2022–23.

Data is sourced from DES (Ministry of Agriculture), ICAR-IIPR, NFSM-Pulses, and the Directorate of Pulses Development. The study identifies growth trajectories, inter-crop comparisons, key constraints, and government initiatives. Visualization is performed using Python/Matplotlib.

- Bengal Gram
- Pigeon Pea
- Lentil
- Green Gram
- Black Gram
- Horse Gram

Review of Literature

Key Studies & Data Sources on Indian Pulse Production

1

DES, MoAFW, GoI (2022)

Area, Production and Yield of Major Pulses – national time-series data used as primary dataset for trend analysis.

2

ICAR-IIPR Annual Report (2021–22)

Variety releases, crop performance benchmarks, and research contributions to productivity improvement.

3

NFSM-Pulses Progress Reports

Scheme-level data on demonstrations, seed distribution, and farm machinery support since 2007.

4

Anil Kumar et al. (2019)

Growth and Instability in Area, Production and Productivity of Major Pulses in Karnataka, Journal of Farm Sciences 32(2).

5

Sharma et al. (2018)

Pulse production in India: Challenges and strategies. Indian Journal of Agronomy, 63 (Supplement): 1–16.

6

Singh et al. (2019) & Kumar & Mishra (2020)

Chickpea production constraints; Pigeon pea yield gaps in Karnataka – regional productivity analysis.

Introduction & Nutritional Importance

Pulses – Backbone of Indian Agriculture & Nutrition

What Are Pulses?

Pulses are annual leguminous crops yielding dried seeds within pods, cultivated in India since the Indus Valley civilization (~2500 BCE). They form the backbone of dietary protein for India's predominantly vegetarian population of 1.4 billion.

- Fix 80–100 kg N/ha via symbiotic Rhizobium
- Reduce dependence on synthetic fertilizers
- Improve soil physical structure & organic matter
- Grown across Kharif, Rabi, and Summer seasons

Nutritional Profile

Protein	20–25%
Dietary Fiber	High
Iron & Zinc	Rich
Folate (B9)	
Complex Carbs	High

Materials & Methods

Data Sources, Analytical Framework & Tools Used

DES, MoAFW

National area, production, productivity time series (2016–2023)

ICAR-IIPR

Variety releases, research data, crop performance benchmarks

NFSM Reports

Scheme-level data, demonstration outcomes since 2007

Karnataka Agri Stats

State-level disaggregated crop data (2016–2021)

Analytical Methods

- **Descriptive Statistics**
- **Time-Series Trend Analysis**
- **CAGR Computation**
- **State-wise Comparative Analysis**
- **Python / Matplotlib**

System Architecture & Data Pipeline

End-to-End Research Workflow Diagram

Data Sources
DES · ICAR
NFSM · APEDA

Data Ingestion
Manual verify
Uniform units

Storage & Processing
Python lists
& arrays

Analysis Engine
NumPy · CAGR
Descriptive stats

Visualization Layer
Matplotlib
150 DPI

Output Layer

- Trend Charts (7 years)
- State-wise Analysis
- Crop Comparison Tables
- Policy Recommendations
- DOCX Final Report

Figure 1: End-to-end data pipeline architecture for Pulse Production Analysis System

Flowchart – Methodology

Step-by-Step Research Process Flow Diagram

START

Research Initiation

Data Collection

DES, ICAR, NFSM, APEDA

Data Cleaning

Manual verify · Uniform units

Data Storage

Python lists & arrays

Statistical Analysis

CAGR · Descriptive · Trend

Visualization

Matplotlib 150 DPI charts

Report Generation

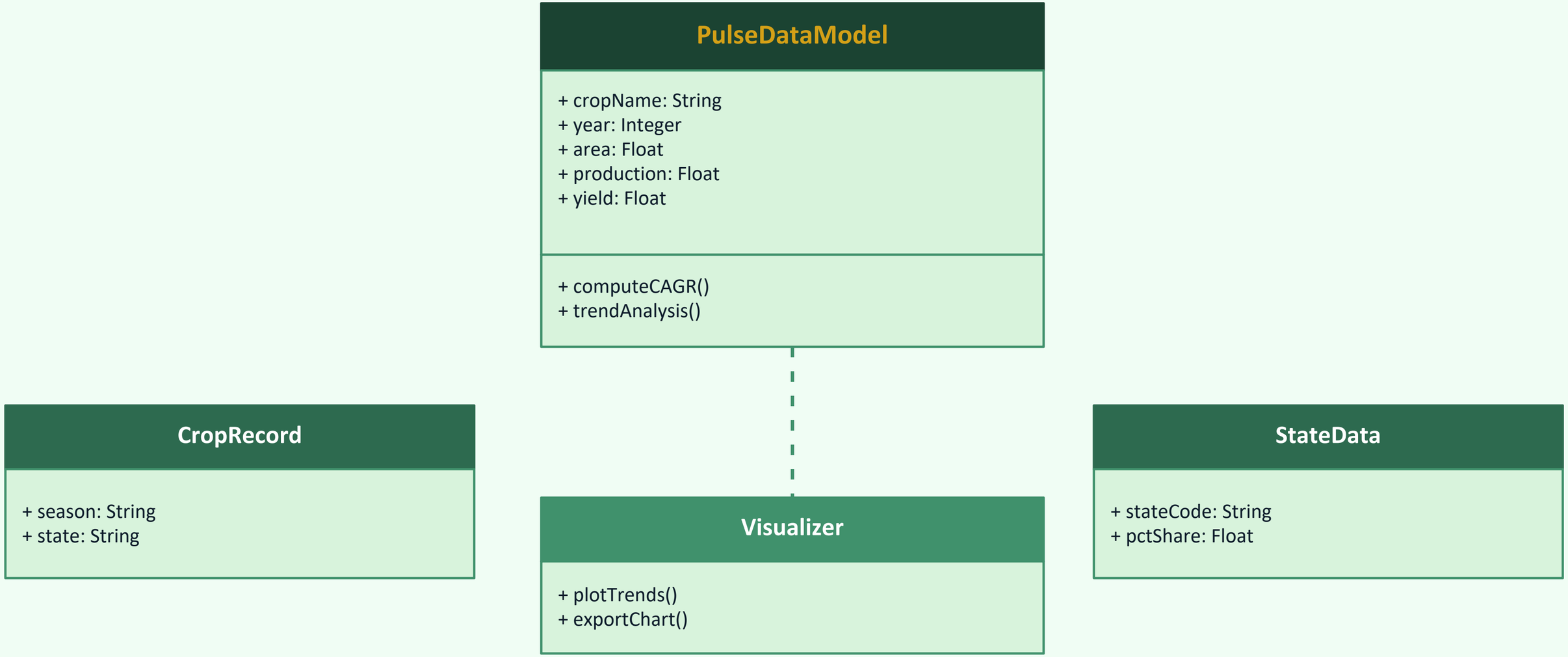
Word/DOCX Final Report

END

Findings Presented

Class Diagram – Pulse Data Model

Object-Oriented Model for Pulse Production Analysis System



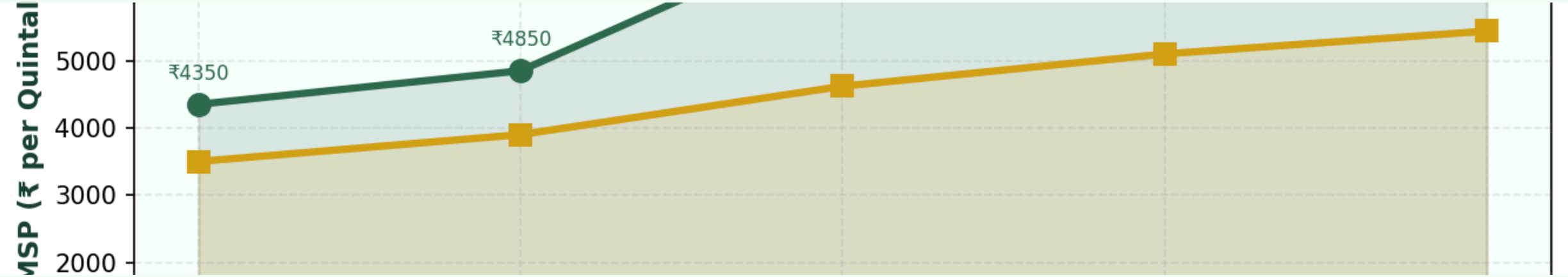
National Pulse Trends – Results

Area, Production & Productivity (2016–2023)

Key Finding: Production grew 21% (23.13 → 28.00 MT) while productivity rose 22.6% (771 → 945 kg/ha) — driven by yield gains, not area expansion.

Results – Comparison with Existing System

Current vs. Traditional Pulse Production Approach



Experiment & Results – Field Photographs

Visual Documentation of Six Major Pulse Crops

Flowering Stage

Bengal Gram (Chickpea)

Pod Formation

Pigeon Pea (Tur)

Maturity Stage

Black Gram (Urad)

Field View

Green Gram (Moong)

Harvesting

Lentil (Masoor)

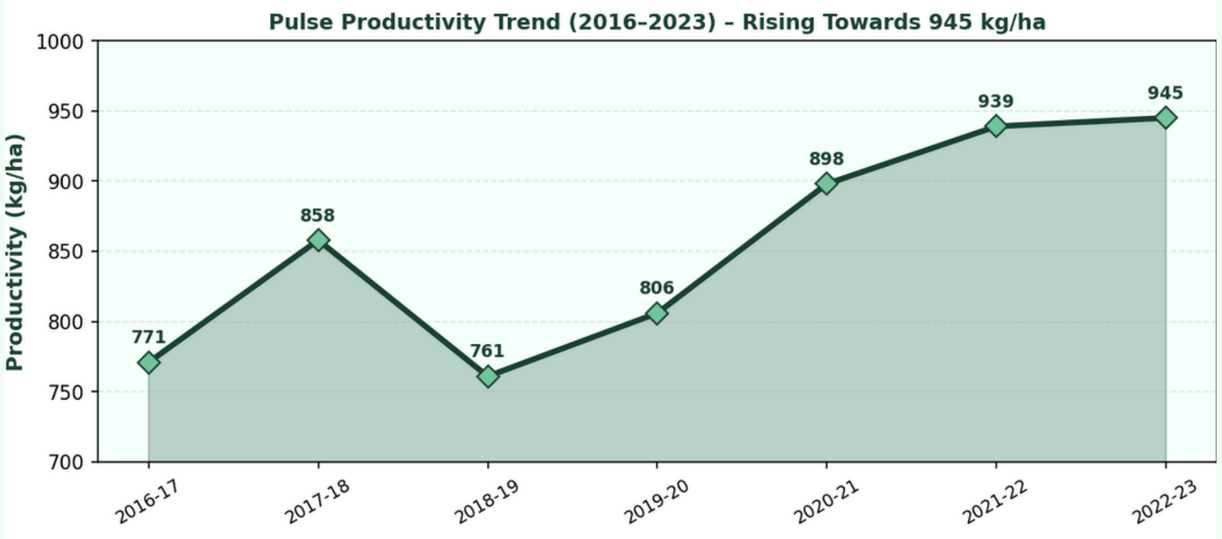
Dried Pods

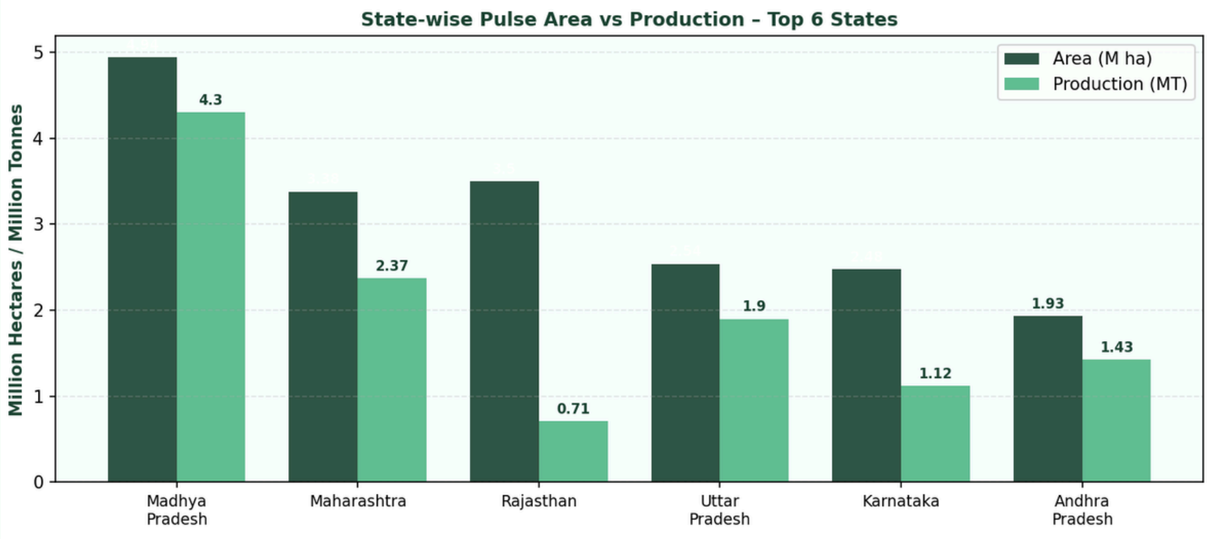
Horse Gram (Kulthi)

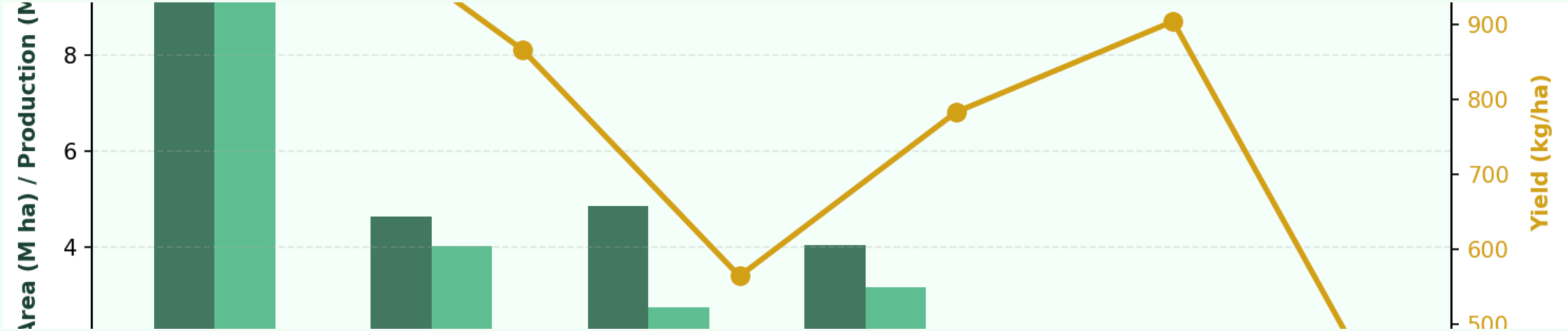
Source: ICAR-IIPR, DES field documentation | Note: Photograph panels illustrate representative crop growth stages

Result & Discussion – Data Visualization

Matplotlib-Generated Charts: Productivity, Six Pulses & State Comparison







Summary & Conclusion

Key Findings & Takeaways from the Study

Production Growth

21% increase (23.13 → 28 MT). Technology-led progress, not area expansion, drove output gains.

Chickpea Dominance

Near-doubling of chickpea output makes it 35%+ of total pulse production, exposing sector to climate risks.

State Disparities

Bihar: 836 kg/ha vs. Rajasthan: 204 kg/ha — agro-climatic and institutional factors drive vast differences.

Karnataka Decline

44% area reduction (2017–18 to 2020–21) due to irrigation-enabled shift to high-value crops.

Productivity Ceiling

945 kg/ha well below global best of 2,000–2,500 kg/ha. This gap is the largest opportunity for improvement.

Conclusion: India's pulse sector shows strong progress but a productivity gap of 1,000+ kg/ha below global best remains the key strategic challenge.

Future Scope & References

Emerging Technologies & Key Data Sources

Future Scope

Precision Agri

AI, IoT & remote sensing for crop management

Genomics & MAS

Marker-assisted breeding for drought & disease tolerance

Biofortification

Enhanced iron, zinc & lysine varieties

Digital Extension

Mobile advisory & AI chatbots in regional languages

Climate Research

Crop models for elevated CO_2 and changing rainfall

Microbiome Eng.

Optimize rhizosphere for N-fixation efficiency

References

[1] DES, MoAFW, GoI – Area, Production & Yield of Pulses (2022)

[2] ICAR-IIPR – Annual Progress Report 2021–22

[3] NFSM-Pulses Progress Reports, MoAFW (2022)

[4] APEDA – Pulses Trade Statistics, apeda.gov.in

[5] Anil Kumar et al. (2019), J. Farm Sciences 32(2)

[6] Sharma et al. (2018), Indian J. Agronomy 63

[7] Singh et al. (2019); Kumar & Mishra (2020)

[8] Kappec.Karnataka.gov.in – State Pulse Stats

THANK YOU

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Questions & Feedback Welcome